Dimension-independent stability of the real Böttcher–Wenzel inequality

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Abstract

The Böttcher–Wenzel inequality bounds the squared Frobenius norm of the commutator of two Frobenius-unit matrices by two. Its real equality cases consist of pairs supported on a common two-dimensional subspace, with traceless, orthogonal restrictions. We give a direct proof that a real pair whose squared commutator norm is $2 - \delta$ lies within $C\sqrt{\delta}$ of such an equality pair, where C is independent of the matrix size. The distance uses the product Frobenius norm, neither matrix is required to be symmetric or normal, and no smallness assumption is imposed on the deficit. The exponent $1/2$ is optimal, already in dimension three. The proof begins with Audenaert's singular-value refinement, which provides a common subspace of dimension at most eight on which both matrices can be compressed. The retained commutator is orthogonal to the terms linear in the compression errors, so the normalized compressed pair has deficit of order δ. The remaining fixed-dimensional estimate follows from an explicit second-variation calculation: the deficit grows quadratically in directions transverse to the equality set. Compactness completes this estimate, and the compression argument combines only the constants for dimensions two through eight. The resulting universal constant is existential.

Keywords: Matrix commutator; Frobenius norm; Böttcher–Wenzel inequality; quantitative stability; equality cases.

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1 Introduction

For real square matrices A and B, the Böttcher–Wenzel inequality is

$$\|[A, B]\|_F^2 \le 2\|A\|_F^2\|B\|_F^2, \qquad [A, B] = AB - BA, \tag{1.1}$$

where $\langle A, B\rangle_F = \operatorname{tr}(A^T B)$ and $\|A\|_F^2 = \langle A, A\rangle_F$; see Böttcher–Wenzel [2]. For matrices of unit Frobenius norm, Cheng–Vong–Wenzel [4] characterize equality by a common two-dimensional supporting subspace. More precisely, equality holds if and only if there is a rank-two orthogonal projection P such that $A = PAP$, $B = PBP$, and the two restrictions to range P are traceless and Frobenius-orthogonal.

We study pairs for which the left-hand side of (1.1) is close to two. The question is whether both matrices must then be close to an equality pair, with an error bound that does not deteriorate as their size increases. For $n \geq 2$, put

$$\delta(A,B) = 2 - \|[A,B]\|_F^2,$$
$$\mathcal{E}_n = \{(A,B) \in (\mathbb{R}^{n\times n})^2 : \|A\|_F = \|B\|_F = 1,\ \delta(A,B) = 0\}.$$

The deficit $\delta(A,B)$ lies in $[0,2]$ for a unit pair. Distances between pairs use the product Frobenius norm,

$$\|(U,V)\|^2 = \|U\|_F^2 + \|V\|_F^2.$$

Theorem 1.1. *There is a finite constant C, independent of $n \geq 2$, such that every pair of real $n \times n$ matrices with $\|A\|_F = \|B\|_F = 1$ satisfies*

$$\operatorname{dist}((A,B),\mathcal{E}_n)^2 \leq C\,\delta(A,B). \tag{1.2}$$

The exponent $1/2$ for distance, equivalently exponent one for squared distance, cannot be increased, even with the dimension fixed at three.

Thus there are matrices A_0, B_0 supported on the same plane, satisfying equality in (1.1), for which $\|A - A_0\|_F^2 + \|B - B_0\|_F^2 \leq C\delta(A,B)$. Separate rank-two approximations of A and B would not give this common support or the equality relation between the two restrictions. Theorem 1.1 controls these requirements simultaneously for unrestricted real pairs. The proof gives existence of a universal constant, not a numerical or optimal value.

For a concrete equality pair, take

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad S = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{1.3}$$

Both matrices have unit Frobenius norm, and their commutator is $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, whose squared norm is two. Extending both by zero gives an equality pair in every dimension. The theorem says that near equality forces a nearby pair with a common two-dimensional support, even when neither input matrix is symmetric. The supporting plane and the two matrices on it are allowed to vary.

1.1 Related bounds and the stability question

Two established refinements supply information about a nearly maximizing pair. Audenaert [1, Corollary 5 in the arXiv version] proves

$$\|[X,Y]\|_F^2 \leq 2\|Y\|_F^2(\sigma_1(X)^2 + \sigma_2(X)^2), \tag{1.4}$$

where the singular values are in decreasing order. For unit A, B, this bounds each squared singular-value tail beyond rank two by $\delta/2$. The Gram-determinant refinement, recorded in Cheng–Jin–Vong [3, Theorem 3.4], also gives $|\langle A,B\rangle_F|^2 \leq \delta/2$. These estimates control rank and angle; the remaining task is to approximate the prescribed pair by an element of $\mathcal{E}_n$ with a constant independent of n.

There are simpler stability arguments under additional assumptions. When one matrix is real symmetric, diagonalization and the entrywise commutator formula give the dimension-independent estimate recorded in Appendix A. Related diagonal and rank-one calculations occur in the weighted setting of Mayumi–Kimura–Ohno–Chruściński [7, Appendix A]. In

A Curvature Gap for Minimal Submanifolds in Spheres (arXiv:2608.16095v1, Lemma 4.1), F. Li states a dimension-independent distance estimate for tuples of real symmetric trace-free matrices of size $n \geq 3$; its two-matrix specialization covers that restricted BW case. Here the cited statement is in arXiv:2608.16095v1, dated 17 August 2026; it is absent from version 3. We claim no novelty for these restricted deductions.

The equality classification also contains estimates useful for stability. For example, Cheng–Vong–Wenzel [4, Lemma 2.2(a)] control unsupported blocks when one matrix already has exact two-dimensional support. Spectral approaches to the inequality and its equality cases appear in Ge–Li–Lu–Zhou [5, Section 3], while Ge–Li–Zhou [6, Section 2] treat related inequalities for several matrices. Our estimate allows errors in both matrices and obtains their common support as part of the conclusion.

In his preprint on sharp commutator bounds on complex simple Lie algebras, D. J. F. Fox (arXiv:2610.11800v2, 11 October 2026; Lemma 4.7, Theorem 6.7 and Corollary 6.10 in the version 2 PDF) proves nondegeneracy of the maximizing manifold for the Cartan-form objective on complex simple Lie algebras. For $\mathfrak{sl}(n, \mathbb{C})$, pullback to matrix pairs and restriction to real matrices give the fixed-dimensional nondegeneracy used below. His spectral formulas also give a normal Hessian gap independent of n after Frobenius normalization. Together with rank-two concentration, uniform Taylor estimates and a compactness reduction, that gap provides another proof of the stability conclusion in Theorem 1.1, which is not stated in that form in Fox's preprint. The contribution of the present proof is its direct treatment in real matrix coordinates and its control of the deficit under compression. Its dimensional reduction preserves a deficit of order δ, so it requires nondegeneracy only in the finitely many dimensions two through eight. We retain the local calculation to make this proof self-contained beyond the established inequality, equality classification and singular-value refinement.

1.2 Proof structure

The proof has two tasks: reduce the ambient dimension without losing the order of the deficit, and prove quadratic growth away from equality in a fixed dimension. We begin with the reduction in Section 2. The row and column spaces of rank-two approximations to both matrices span a common space of dimension at most eight. Compression to that space discards matrices of norm $O(\sqrt{\delta})$. Their linear contribution to the commutator is orthogonal to the retained commutator, as the block diagram in Figure 1 shows. The normalized compressed pair therefore still has deficit $O(\delta)$.

Once this reduction is established, only stability in dimensions two through eight remains. Sections 3–6 prove the required fixed-dimensional bound. We first orthogonalize the pair, then calculate the second variation at equality. Its vanishing directions correspond exactly to changes of the supporting plane and of the orthonormal pair within that plane. The deficit is thus positive to second order in every remaining tangent direction. Compactness turns this local fact into a global bound in each fixed dimension; the argument is a standard consequence of a nondegenerate minimum along a manifold [8, Proposition 2.3], supplied here for the specific constraints and distance. Section 7 proves that the distance exponent cannot be improved.

2 Reduction to a bounded dimension

We first show why fixed-dimensional stability will suffice. Suppose that, for each $m \geq 2$, there is a finite constant C_m such that

$$\operatorname{dist}((X,Y),\mathcal{E}_m)^2 \leq C_m \delta(X,Y) \qquad \text{whenever } \|X\|_F = \|Y\|_F = 1.$$

Lemma 6.1 will establish this assertion in Section 6. Here we prove that the constants for $2 \leq m \leq 8$ imply Theorem 1.1's stability bound for every n. The issue is quantitative: an error of order $\sqrt{\delta}$ in the matrices must produce an error of order δ in the deficit.

Let $\|A\|_F = \|B\|_F = 1$ and write $\delta = \delta(A,B)$. Audenaert's refinement (1.4), applied in both orders, gives

$$\sum_{j\geq 3} \sigma_j(A)^2 \leq \frac{\delta}{2}, \qquad \sum_{j\geq 3} \sigma_j(B)^2 \leq \frac{\delta}{2}. \tag{2.1}$$

Choose rank-at-most-two singular-value truncations A_2, B_2. Thus $\|A - A_2\|_F^2 \leq \delta/2$ and $\|B - B_2\|_F^2 \leq \delta/2$. Take W to contain

$$\operatorname{ran} A_2 + \operatorname{ran} A_2^T + \operatorname{ran} B_2 + \operatorname{ran} B_2^T,$$

enlarging this span if necessary to have $2 \leq m = \dim W \leq \min\{8, n\}$. Each of the four spaces has dimension at most two. Including both the row and column spaces makes the same orthogonal projection P satisfy $PA_2P = A_2$ and $PB_2P = B_2$.

Set

$$a = PAP, \qquad b = PBP, \qquad e = A - a, \qquad f = B - b, \qquad x = \|e\|_F^2, \quad y = \|f\|_F^2, \quad t = x + y.$$

Since $X \mapsto PXP$ is the Frobenius-orthogonal projection onto the matrices supported on W, it gives no larger approximation error than A_2 or B_2. Consequently

$$x, y \leq \frac{\delta}{2}, \qquad t \leq \delta, \qquad \|a\|_F^2 = 1 - x, \quad \|b\|_F^2 = 1 - y. \tag{2.2}$$

Lemma 2.1. *With the preceding notation, suppose that $\delta < 1/2$. Then the normalized compressions $\widehat{a} = a/\|a\|_F$ and $\widehat{b} = b/\|b\|_F$ are defined and satisfy*

$$\delta(\widehat{a}, \widehat{b}) \leq \delta + 4t \leq 5\delta.$$

Proof. Decompose the commutator as

$$Z = [A,B] = C + L + F, \qquad C = [a,b], \quad L = [a,f] + [e,b], \quad F = [e,f].$$

Relative to $W \oplus W^{\perp}$, the retained matrices have only their upper-left blocks, denoted by a_{11}, b_{11}, whereas $e_{11} = f_{11} = 0$. The commutator C is supported on W, whereas L has only the two off-diagonal blocks. Figure 1 displays the block products and their disjoint positions. In particular $\langle C, L\rangle_F = 0$. This removes the term linear in the discarded matrices from the squared commutator norm. The surviving contributions are $\|L\|_F^2$ and the terms containing the quadratic error F.

Applying Cauchy–Schwarz and Frobenius submultiplicativity to the two blocks of L gives

$$\begin{aligned} \|L\|_F^2 &\leq (\|a\|_F^2 + \|b\|_F^2)(\|e_{12}\|_F^2 + \|f_{12}\|_F^2 + \|e_{21}\|_F^2 + \|f_{21}\|_F^2) \\ &\leq (2 - t)t \leq 2t. \end{aligned}$$

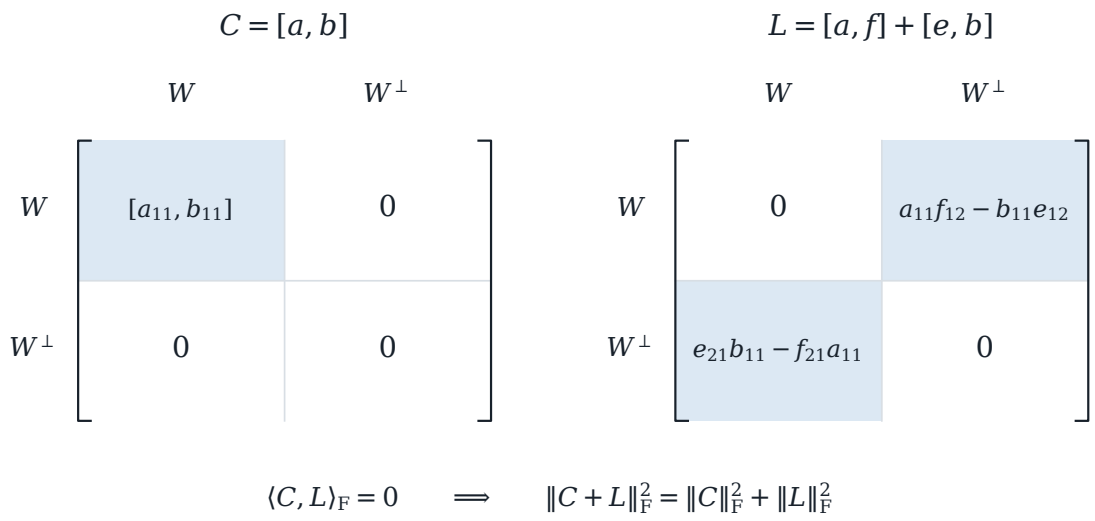


Figure 1: Block supports relative to $W \oplus W^\perp$. The retained commutator $C = [a, b]$ and the linear error $L = [a, f] + [e, b]$ occupy disjoint blocks, so their Frobenius inner product is zero. Block areas do not represent dimensions, and the external blocks are absent when $W = \mathbb{R}^n$.

Inequality (1.1) gives $\|F\|_F \le \sqrt{2xy}$ and $\|Z\|_F \le \sqrt{2}$. Using $C \perp L$ and $Z - F = C + L$, we therefore obtain

$$\begin{aligned}\|Z\|_F^2 - \|C\|_F^2 &= \|L\|_F^2 + 2\langle Z, F\rangle_F - \|F\|_F^2 \\ &\le 2t + 4\sqrt{xy} \le 4t.\end{aligned}$$

Hence $2 - \|C\|_F^2 \le \delta + 4t$. By (2.2), $1 - x$ and $1 - y$ are positive and at most one. Normalization can only increase the squared commutator norm, so

$$\delta(\widehat{a}, \widehat{b}) = 2 - \frac{\|C\|_F^2}{(1-x)(1-y)} \le 2 - \|C\|_F^2 \le \delta + 4t \le 5\delta.$$

□

The orthogonality in this proof is what preserves the order of the deficit: $\|L\|_F = O(\sqrt{\delta})$ enters through its square, while $\|F\|_F = O(\delta)$ may enter linearly. Thus the fixed-dimensional estimate can be applied with a deficit of order δ.

Reduction of Theorem 1.1 to fixed-dimensional stability. Let $C_* = \max_{2 \le m \le 8} C_m$, which is finite by Lemma 6.1. If $\delta < 1/2$, identify the subspace W constructed above isometrically with $\mathbb{R}^m$. The normalized compressions are Frobenius-unit matrices on W, so Lemmas 6.1 and 2.1 give an equality pair (A_0, B_0) on W with

$$\|\widehat{a} - A_0\|_F^2 + \|\widehat{b} - B_0\|_F^2 \le 5C_*\delta.$$

Its zero extension belongs to $\mathcal{E}_n$.

To transfer this approximation to the original pair, put $r = \|a\|_F$. The compression error e

is orthogonal to every matrix supported on W. Since $\|\widehat{a}\|_F = \|A_0\|_F = 1$ and $x = 1 - r^2$,

$$\begin{aligned}\|A - A_0\|_F^2 &= x + \|r\widehat{a} - A_0\|_F^2 \\ &= x + (1-r)^2 + r\|\widehat{a} - A_0\|_F^2 \\ &= 2(1-r) + r\|\widehat{a} - A_0\|_F^2 \\ &\le 2x + \|\widehat{a} - A_0\|_F^2.\end{aligned}$$

Here $0 < r \le 1$ and $1 - r = x/(1+r) \le x$. Applying the same identity to B and using $t \le \delta$ yields

$$\operatorname{dist}((A,B), \mathcal{E}_n)^2 \le 2t + 5C_*\delta \le (2 + 5C_*)\delta.$$

If $\delta \ge 1/2$, choose any equality pair and choose the sign of each of its matrices so that its inner product with the corresponding input is nonnegative. The resulting squared distance is at most $4 \le 8\delta$. Thus $C = \max\{8, 2 + 5C_*\}$ proves (1.2) for every $n \ge 2$, provided Lemma 6.1 holds. □

The ambient dimension has now left the problem. We next establish Lemma 6.1 by examining equality pairs and the quadratic change of the deficit near them. The argument works in every fixed dimension, although the reduction above uses only sizes two through eight.

3 Reduction to orthonormal pairs

The local calculation is simplest when the two unit matrices are also orthogonal for the Frobenius inner product. Near equality already forces their inner product to be small. The following consequence of the Gram-determinant refinement [3, Theorem 3.4] shows that orthogonalization has a controlled cost and does not increase the deficit.

Lemma 3.1. *Let $\|A\|_F = \|B\|_F = 1$ and $\alpha = \langle A, B\rangle_F$. Then $\alpha^2 \le \delta(A,B)/2$. If $|\alpha| < 1$, set*

$$B' = \frac{B - \alpha A}{\sqrt{1 - \alpha^2}}.$$

The pair (A, B') is orthonormal and

$$\delta(A, B') = \frac{\delta(A,B) - 2\alpha^2}{1 - \alpha^2} \le \delta(A,B), \qquad \|B - B'\|_F^2 \le 2\alpha^2 \le \delta(A,B).$$

Proof. Since $[A, B - \alpha A] = [A, B]$, BW gives $2 - \delta \le 2(1 - \alpha^2)$. Also $\langle B, B'\rangle_F = \sqrt{1-\alpha^2}$, whence $\|B - B'\|_F^2 = 2(1 - \sqrt{1-\alpha^2}) \le 2\alpha^2$. The deficit formula follows from the scaling of the commutator. Its upper bound uses $\delta \le 2$. □

Write

$$\mathcal{M}_n = \{(A,B) : \|A\|_F = \|B\|_F = 1,\ \langle A, B\rangle_F = 0\}.$$

This smooth constraint set is the Stiefel manifold of orthonormal two-frames in the Euclidean space $\mathbb{R}^{n\times n}$. Its tangent vectors will be obtained by differentiating its three displayed constraints. An orthogonal change of the two frame vectors preserves $\mathcal{M}_n$ and the commutator norm. It need not preserve two individually normalized vectors outside $\mathcal{M}_n$, which is why Lemma 3.1 precedes the change of frame.

A straight perturbation of a pair in $\mathcal{M}_n$ need not remain in $\mathcal{M}_n$. For Taylor expansion, we therefore extend the deficit to the nonnegative polynomial on all matrix pairs

$$D(A,B) = 2(\|A\|_F^2\|B\|_F^2 - \langle A, B\rangle_F^2) - \|[A,B]\|_F^2. \tag{3.1}$$

It is nonnegative: remove the component of one matrix parallel to the other and apply BW, with zero matrices treated directly. Under an orthogonal change of the pair, the Gram determinant in (3.1) is unchanged and the commutator is multiplied by the determinant of that change. Thus D is invariant. On $\mathcal{M}_n$, $D = \delta$. At an equality pair its ambient gradient vanishes because $D \geq 0$ there and $D = 0$.

4 The equality set and its coordinates

The equality classification [4] describes an equality pair by its common supporting plane and an orthonormal pair of traceless matrices on that plane. We first show that these parameters give a smooth embedded submanifold of $\mathcal{M}_n$. This will identify the directions along which the deficit vanishes before we calculate its second variation in the remaining directions.

If X is a real traceless 2×2 matrix, multiplication gives

$$XX^T + X^T X = \|X\|_F^2 I_2.$$

Consequently an equality pair determines the orthogonal projection onto its support by

$$P = \tfrac{1}{2}(AA^T + A^T A + BB^T + B^T B). \tag{4.1}$$

The support is therefore unique, even when one of the matrices has rank one.

Locally, a two-dimensional subspace W of $\mathbb{R}^n$ is the graph of a linear map from a fixed plane to its orthogonal complement, so it has $2(n-2)$ coordinates. Choose an orthonormal basis of W smoothly in these local coordinates. In that basis, traceless endomorphisms of W form a three-dimensional Euclidean space, and equality pairs are its orthonormal two-frames. This frame space is $V_2(\mathbb{R}^3)$ and has dimension three. Extending the two matrices by zero gives smooth local parameterizations of $\mathcal{E}_n$. They have smooth inverses: recover W from (4.1), then restrict A and B to the chosen basis of W. Thus $\mathcal{E}_n$ is an embedded submanifold of $\mathcal{M}_n$, of dimension

$$\dim \mathcal{E}_n = 2(n-2) + 3 = 2n - 1. \tag{4.2}$$

It is compact because it is the zero set of δ in the compact manifold $\mathcal{M}_n$.

Two orthogonal operations will be used to choose coordinates for the second variation. An orthogonal change of the pair is

$$(A, B) \longmapsto (r_{11}A + r_{12}B,\ r_{21}A + r_{22}B), \qquad R = (r_{ij}) \in O(2),$$

whereas simultaneous orthogonal conjugation is

$$(A, B) \longmapsto (OAO^T, OBO^T), \qquad O \in O(n).$$

The first operation multiplies the commutator by $\det R$; the second sends it to $O[A, B]O^T$. Both are linear isometries of the space of matrix pairs, preserve $\mathcal{M}_n$ and D, and carry $\mathcal{E}_n$ to itself. They therefore preserve the constrained second variation.

Retain H, S from (1.3) and put

$$K = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

These three matrices form an orthonormal basis of the traceless real 2×2 matrices. The first two are symmetric and the third is skew-symmetric. The two-dimensional span of an

equality pair on its support meets the two-dimensional space of symmetric traceless matrices nontrivially. A change of the pair makes a unit vector in this intersection the first matrix. Orthogonal conjugation on the support diagonalizes that matrix as H. The second matrix, being traceless, unit, and orthogonal to H, then lies in the span of S, K. We can consequently calculate at

$$A = H \oplus 0, \qquad B = B_2 \oplus 0, \qquad B_2 = cS + sK, \quad c^2 + s^2 = 1. \tag{4.3}$$

In particular, $c = s = 1/\sqrt{2}$ gives $B_2 = E_{12}$; the normal form includes rank-one matrices without a separate case.

5 The second variation at equality

Moving within $\mathcal{E}_n$ keeps the deficit zero. To prove a quadratic distance estimate, we must show that no other tangent direction has zero second variation. We separate changes within the supporting plane, changes wholly outside it, and changes between the plane and its orthogonal complement. This decomposition also identifies the geometric meaning of the kernel at the end of the calculation.

Lemma 5.1. *At every $q \in \mathcal{E}_n$, the Hessian of $\delta|_{\mathcal{M}_n}$ is positive semidefinite and its kernel is exactly $T_q\mathcal{E}_n$.*

Proof. The quadratic term. For $(U, V) \in T_{(A,B)}\mathcal{M}_n$, differentiating the three defining constraints gives

$$\langle A, U\rangle_F = 0, \qquad \langle B, V\rangle_F = 0, \qquad \langle A, V\rangle_F + \langle B, U\rangle_F = 0. \tag{5.1}$$

The commutator has the exact expansion

$$[A + tU, B + tV] = [A, B] + t([U, B] + [A, V]) + t^2[U, V].$$

Under (5.1), the coefficient of t^2 in the Gram determinant

$$\|A + tU\|_F^2 \|B + tV\|_F^2 - \langle A + tU, B + tV\rangle_F^2$$

is $\|U\|_F^2 + \|V\|_F^2$. Thus

$$D(A + tU, B + tV) = t^2 Q(U, V) + O(t^3),$$

where

$$Q(U, V) = 2(\|U\|_F^2 + \|V\|_F^2) - \|[U, B] + [A, V]\|_F^2 - 2\langle [A, B], [U, V]\rangle_F. \tag{5.2}$$

Since the ambient gradient of D vanishes on $\mathcal{E}_n$, its Hessian restricted to $T_{(A,B)}\mathcal{M}_n$ is the Hessian of $\delta|_{\mathcal{M}_n}$. The latter quadratic form is therefore $2Q$.

Block decomposition. Use (4.3) and write the variations relative to the common support and its orthogonal complement as

$$U = \begin{pmatrix} U_0 & X \\ Y^T & U_e \end{pmatrix}, \qquad V = \begin{pmatrix} V_0 & Z \\ T^T & V_e \end{pmatrix},$$

where U_0, V_0 are 2×2, U_e, V_e are $(n-2) \times (n-2)$, and X, Y, Z, T are $2 \times (n-2)$ matrices. For $n = 2$, all external blocks are absent. The block products relevant to (5.2) are

$$[U, B] + [A, V] = \begin{pmatrix} [U_0, B_2] + [H, V_0] & HZ - B_2 X \\ Y^T B_2 - T^T H & 0 \end{pmatrix}, \tag{5.3}$$

$$[U, V]_{11} = [U_0, V_0] + XT^T - ZY^T. \tag{5.4}$$

Because $[A, B]$ is supported on the first block, these formulas show that Q is the sum of an internal contribution, an external contribution, and one off-diagonal contribution for each external coordinate. There are no mixed terms between these groups.

Changes within and outside the support. For the internal contribution, write $N = -sS + cK$ and $J = I_2/\sqrt{2}$. The tangent constraints give

$$U_0 = \alpha J + \xi B_2 + \eta N, \qquad V_0 = \beta J - \xi H + \zeta N.$$

For traceless real 2×2 matrices X_0, Y_0, the identity [4, Aspect 3]

$$\|[X_0, Y_0]\|_F^2 = 2(\|X_0\|_F^2 \|Y_0\|_F^2 - \langle X_0, Y_0 \rangle_F^2)$$

follows by expansion in H, S, K, using $[H, S] = \sqrt{2}K$, $[H, K] = \sqrt{2}S$, and $[S, K] = -\sqrt{2}H$. Thus D vanishes identically on pairs of traceless 2×2 matrices, and the traceless internal contribution to its Hessian is zero. The scalar terms commute with all internal matrices and are orthogonal to the traceless terms. Their contribution to Q is $2(\alpha^2 + \beta^2)$. The internal kernel therefore consists of the three parameters ξ, η, ζ, which are the tangent directions of the orthonormal frame on the fixed support. By (5.3)–(5.4), the external contribution is $2(\|U_e\|_F^2 + \|V_e\|_F^2)$.

Changes between the support and its complement. For the off-diagonal contribution, set $u = c + s$ and $v = c - s$, so that

$$B_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & u \\ v & 0 \end{pmatrix}, \qquad [H, B_2] = \begin{pmatrix} 0 & u \\ -v & 0 \end{pmatrix}, \qquad u^2 + v^2 = 2.$$

For one external coordinate, let $x, y, z, w \in \mathbb{R}^2$ be the corresponding columns of X, Y, Z, T. The two off-diagonal blocks in (5.3) are

$$Hz - B_2 x = \frac{1}{\sqrt{2}} \begin{pmatrix} z_1 - ux_2 \\ -z_2 - vx_1 \end{pmatrix}, \qquad y^T B_2 - w^T H = \frac{1}{\sqrt{2}} \begin{pmatrix} vy_2 - w_1 & uy_1 + w_2 \end{pmatrix}.$$

The contribution of (5.4) to the last term in (5.2) is determined by

$$\langle [H, B_2], xw^T - zy^T \rangle_F = ux_1 w_2 - vx_2 w_1 - uz_1 y_2 + vz_2 y_1.$$

Substitution gives

$$\begin{aligned} Q_{\text{cross}} = {} & 2(|x|^2 + |y|^2 + |z|^2 + |w|^2) \\ & - \tfrac{1}{2}\left[(z_1 - ux_2)^2 + (z_2 + vx_1)^2 + (vy_2 - w_1)^2 + (uy_1 + w_2)^2\right] \\ & - 2(ux_1 w_2 - vx_2 w_1 - uz_1 y_2 + vz_2 y_1). \end{aligned}$$

The variables split into two groups. For $(a,b,p,q) = (x_1, y_1, z_2, w_2)$, the contribution is

$$\begin{aligned} Q_1 &= 2(a^2+b^2+p^2+q^2) - \tfrac{1}{2}\big[(p+va)^2 + (q+ub)^2\big] - 2(uaq+vbp) \\ &= \frac{3}{2}\left(p - \frac{v}{3}(a+2b)\right)^2 + \frac{3}{2}\left(q - \frac{u}{3}(2a+b)\right)^2 + \frac{2}{3}(a-b)^2. \end{aligned}$$

For $(a,b,p,q) = (x_2, y_2, z_1, w_1)$, it is

$$\begin{aligned} Q_2 &= 2(a^2+b^2+p^2+q^2) - \tfrac{1}{2}\big[(p-ua)^2 + (q-vb)^2\big] + 2(vaq+ubp) \\ &= \frac{3}{2}\left(p + \frac{u}{3}(a+2b)\right)^2 + \frac{3}{2}\left(q + \frac{v}{3}(2a+b)\right)^2 + \frac{2}{3}(a-b)^2. \end{aligned}$$

In both completions, $u^2+v^2 = 2$ gives the coefficient $2/3$ in the last square. Hence $Q_{\mathrm{cross}} = Q_1 + Q_2$ is nonnegative, and its kernel is generated by

$$(x_1, y_1, z_2, w_2) = (1, 1, v, u), \qquad (x_2, y_2, z_1, w_1) = (1, 1, -u, -v),$$

with the variables in the other group zero. No division by u or v has been used, so these conclusions include their zero values.

Identification of the kernel. These kernel vectors arise from rotations of the supporting plane. On the sum of that plane and the chosen external coordinate, let

$$\Omega_k = \begin{pmatrix} 0_{2\times 2} & -k \\ k^T & 0 \end{pmatrix}, \qquad k \in \mathbb{R}^2.$$

Differentiate simultaneous conjugation by $\exp(t\Omega_k)$ at $t = 0$. It gives $(U, V) = ([\Omega_k, A], [\Omega_k, B])$, with

$$x = y = Hk, \qquad z = B_2 k, \qquad w = B_2^T k.$$

The choices $k = (\sqrt{2}, 0)^T$ and $k = (0, -\sqrt{2})^T$ give the two displayed generators. The three internal frame directions and these two support directions for each of the $n-2$ external coordinates account for the whole kernel of Q. They are tangent to $\mathcal{E}_n$ and have total dimension $3 + 2(n-2) = \dim \mathcal{E}_n$. Thus the Hessian is positive semidefinite and its kernel is $T_{(A,B)}\mathcal{E}_n$. □

6 A fixed-dimensional stability bound

The Hessian calculation implies quadratic growth of the deficit away from the equality family. The following compactness argument supplies the global estimate in each fixed dimension. It is the usual consequence of a Morse–Bott minimum; see also Rebjock–Boumal [8, Proposition 2.3].

Lemma 6.1. *For each fixed $m \geq 2$ there is $C_m < \infty$ such that* $\operatorname{dist}((A,B), \mathcal{E}_m)^2 \leq C_m \delta(A,B)$ *for all individually normalized real $m \times m$ pairs.*

Proof. First restrict to $\mathcal{M}_m$. If no constant works there, we can choose $z_j \in \mathcal{M}_m \setminus \mathcal{E}_m$ such that

$$\frac{D(z_j)}{\operatorname{dist}(z_j, \mathcal{E}_m)^2} \longrightarrow 0.$$

The distances are bounded because $\mathcal{M}_m$ is compact, so $D(z_j) \to 0$. After taking a subsequence, z_j converges to a zero of $D|_{\mathcal{M}_m}$, and hence to an equality pair. Choose a nearest $q_j \in \mathcal{E}_m$ and write

$$t_j = \|z_j - q_j\| \longrightarrow 0, \qquad v_j = \frac{z_j - q_j}{t_j}.$$

A further subsequence satisfies $q_j \to q \in \mathcal{E}_m$ and $v_j \to v$, where $\|v\| = 1$.

The limit v is tangent to $\mathcal{M}_m$. Indeed, writing $q_j = (A_j, B_j)$ and $v_j = (U_j, V_j)$, the constraints for q_j and $q_j + t_j v_j = z_j$ give

$$\begin{aligned} 2\langle A_j, U_j\rangle_F + t_j\|U_j\|_F^2 &= 0, \\ 2\langle B_j, V_j\rangle_F + t_j\|V_j\|_F^2 &= 0, \\ \langle A_j, V_j\rangle_F + \langle B_j, U_j\rangle_F + t_j\langle U_j, V_j\rangle_F &= 0. \end{aligned}$$

Their limits are the tangent constraints (5.1) at q. Nearest-point minimality also gives $v_j \perp T_{q_j}\mathcal{E}_m$, by differentiating squared distance along any smooth curve in $\mathcal{E}_m$ through q_j. Since $\mathcal{E}_m$ is smooth, its tangent spaces vary continuously, and $v \perp T_q\mathcal{E}_m$.

Although their limit is tangent to $\mathcal{M}_m$, the secants v_j need not be tangent at q_j. We therefore apply Taylor's formula to the ambient polynomial D. Its value and gradient vanish at q_j, and its third derivatives are bounded in a neighborhood of the compact set $\mathcal{E}_m$. It follows that

$$\frac{D(z_j)}{t_j^2} = \tfrac{1}{2}\nabla^2 D(q_j)[v_j, v_j] + O(t_j) \longrightarrow \tfrac{1}{2}\nabla^2 D(q)[v, v].$$

The limit is strictly positive: v is a unit vector tangent to $\mathcal{M}_m$ and perpendicular to the kernel $T_q\mathcal{E}_m$ identified in Lemma 5.1. This contradicts the defining ratio for z_j. Thus some finite K_m gives the required estimate on $\mathcal{M}_m$.

For an individually normalized pair with $\delta < 1$, apply Lemma 3.1. Its orthonormal pair (A, B') satisfies $\|B - B'\|_F^2 \le \delta$ and $\delta(A, B') \le \delta$. The triangle inequality, followed by $(a+b)^2 \le 2a^2 + 2b^2$, gives

$$\operatorname{dist}((A, B), \mathcal{E}_m)^2 \le 2\|B - B'\|_F^2 + 2\operatorname{dist}((A, B'), \mathcal{E}_m)^2 \le 2(1 + K_m)\delta.$$

For $\delta \ge 1$, fix an equality pair and choose the sign of each matrix so that its inner product with the corresponding input is nonnegative. Each squared distance is then at most two, so $\operatorname{dist}((A, B), \mathcal{E}_m)^2 \le 4$. Consequently $C_m = \max\{4, 2(1 + K_m)\}$ suffices. □

This proves the fixed-dimensional estimate used in Section 2, and therefore completes the stability bound in Theorem 1.1. The next example establishes the optimality of its exponent.

7 Optimality of the distance exponent

In dimension three, embed H, S into the first two coordinates and put

$$A_t = \frac{H + tE_{33}}{\sqrt{1 + t^2}}, \qquad B_t = S, \qquad 0 < t < 1/\sqrt{2}.$$

Both matrices have unit Frobenius norm, and E_{33} commutes with S. It follows that

$$\delta(A_t, B_t) = \frac{2t^2}{1 + t^2}.$$

Every matrix in an equality pair has rank at most two. The smallest singular value of A_t is $t/\sqrt{1 + t^2}$, so its distance to any rank-two matrix is at least that value. Consequently

$$\operatorname{dist}((A_t, B_t), \mathcal{E}_3)^2 \ge \frac{t^2}{1 + t^2} = \frac{\delta(A_t, B_t)}{2}.$$

A bound for distance with exponent strictly greater than $1/2$ would contradict this family as $t \to 0$. This completes the sharpness part of Theorem 1.1.

A The elementary case with one symmetric matrix

This appendix records a restricted consequence of the standard diagonal commutator formula, independently of the geometric proof above. It explains why allowing a nonsymmetric second matrix alone is insufficient to describe the contribution of Theorem 1.1.

Proposition A.1. *If A is real symmetric and $\|A\|_F = \|B\|_F = 1$, then*

$$\operatorname{dist}((A,B),\mathcal{E}_n)^2 \le 256\,\delta(A,B).$$

If $\delta(A,B) \le 1/64$, the right-hand side can be replaced by $3\delta(A,B)$.

Proof. Conjugate orthogonally so that $A = \operatorname{diag}(a_1,\dots,a_n)$, and write $B = (b_{ij})$. The familiar diagonal commutator calculation, also used in the weighted setting in [7, Appendix A], gives

$$\delta = \sum_{i,j} b_{ij}^2 (2 - (a_i - a_j)^2).$$

Assume $\delta \le 1/64$. Since $\sum b_{ij}^2 = 1$, there are distinct indices p, q for which $d = |a_p - a_q| \ge \sqrt{2-\delta}$. Choose the sign of

$$A_0 = \pm \frac{E_{pp} - E_{qq}}{\sqrt{2}}$$

so that $\langle A, A_0\rangle_F = d/\sqrt{2}$. Then

$$\|A - A_0\|_F^2 = 2 - \sqrt{2}d \le 2 - \sqrt{4 - 2\delta} \le \delta.$$

For (i,j) other than (p,q) or (q,p), the corresponding difference between diagonal entries of A_0 has absolute value at most $1/\sqrt{2}$. Consequently

$$|a_i - a_j| \le \frac{1}{\sqrt{2}} + \sqrt{2}\|A - A_0\|_F \le \frac{1}{\sqrt{2}} + \sqrt{2\delta} < 1.$$

Let B_* be the Frobenius projection of B onto $\operatorname{span}\{E_{pq}, E_{qp}\}$ and set $\eta = \|B - B_*\|_F^2$. Every summand in the deficit formula is nonnegative, since $(a_i - a_j)^2 \le 2$. The estimate for the remaining gaps therefore gives $\eta \le \delta$, so $B_* \ne 0$. For $B_0 = B_*/\|B_*\|_F$ we have

$$\|B - B_0\|_F^2 = 2(1 - \sqrt{1-\eta}) \le 2\eta \le 2\delta.$$

The pair (A_0, B_0) belongs to $\mathcal{E}_n$, proving the local bound. For $\delta > 1/64$, the general bound for squared distance by 4, obtained by choosing signs of an equality pair, gives $4 < 256\delta$. □

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